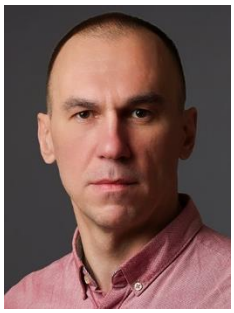
Mikhail Filatov



Contact Information
Central Quad, City Campus, Grangegorman, Dublin 7, Dublin, Ireland
Office: PH-611
Phone: +353-874582449
Email: mikhail.filatov@tudublin.ie
Personal web-site: https://filatov.science/

HIGHER EDUCATION QUALIFICATIONS

2020 **Postgraduate Certificate in University Learning and Teaching**
Learning, Teaching and Technology Centre (LTTC), TU Dublin, Ireland

2005 – 2008 **PhD in Organic Chemistry**
Department of Chemistry, Moscow State University, Moscow, Russia
Thesis title: "General synthetic approach to porphyrins and dipyrrins with a π-extended system". Supervisors: I.P. Beletskaya and A.V. Cheprakov

2000 – 2005 **Diploma of Chemist with a Red Diploma** (distinction) (NFQ equivalent: Level 9)
Department of Chemistry, Moscow State University, Moscow, Russia

CURRENT POSITION

2017 – **Lecturer in Organic Chemistry** (permanent)
School of Chemical and Biopharmaceutical Sciences, Technological University Dublin, Ireland

PREVIOUS POSITIONS

2015 – 2017 **Marie Curie Research Fellow (MSCA-IF)**
School of Chemistry, Trinity College Dublin, Ireland

2014 – 2015 **Senior Researcher**
Institute of Polymers, Bulgarian Academy of Sciences, Sofia, Bulgaria

2010 – 2014 **Postdoctoral Researcher**
Max Planck Institute for Polymer Research, Mainz, Germany

2008 – 2009 **CNRS Postdoctoral Researcher**
Institute of Molecular Chemistry, University of Burgundy, Dijon, France

2008 **Visiting Researcher**
Department of Biochemistry and Biophysics, University of Pennsylvania, Philadelphia, USA

2005 – 2008 **Managing Director**
Esterkem Ltd., startup chemical company, Moscow, Russia

FELLOWSHIPS AND AWARDS (INCLUDING MAJOR GRANTS)

2022 – **Research Ireland Frontiers for the Future Award** (Principal Investigator)
Project: Dyes with Switchable Intersystem Crossing for Photonics (DyeSICPhoto, 21/FFP-A/9214)

2020 – 2024 **TU Dublin Research Scholarship** (Principal Investigator)
Project: Heavy-Atom-Free Photosensitizing Materials

2015 – 2017 **Marie Curie Intra-European Fellowship** (Principal Investigator)
"Controlled Singlet Oxygen Release Sensitizer in Photodynamic Therapy"

2010 – 2014 **Max Planck Society Scholarship** (individual postdoc grant - stipend for outstanding scientists from abroad)
Project: New Functional Dyes for NIR to Visible Light Upconversion

2005 – 2006 **Russian Foundation for Assistance to Small Innovative Enterprises** (grant for establishing a startup company)
Project: Development of Technology of 24-Epibrassinolide Production

PATENTED INNOVATIONS

1. Long-term stable composition, such as phosphorescent composition or TTA-photon upconversion composition, EP 2 851 407 A1, US 2016/0222286 A1, WO 2015/044129 A1, 2015.

2. Method of Synthesis of 5,5'-Disubstituted π-extended Dipyrromethenes and Their Use as Analytical Reagents for Metal Ions and Fluorescent Imaging Probes, US 2011/0144351 A1, 2009.

3. Method of Reduction of Unsaturated Ketones into Saturated Ketones, RU 2 293 720 C1, 2007.

4. Method of Synthesis of 24-Epibrassinolide, RU 2 272 044 C1, 2006.

PUBLICATIONS

Summary: 50 scientific papers published (27 as a corresponding author), 1 book chapter.
h index = 29, > 2900 citations (Google Scholar), 8 papers selected for journal covers

Representative publications

(* corresponding author)

V. N. K. Mamillapalli, I. A. Okkelman, F. Akhssas, G. Sambucari, A. C. Debruyne, D. Beniwal, C. Bucher, M. Di Donato, R. I. Dmitriev, M.A. Filatov*, Hydration-Gated Photoinduced Electron Transfer in BODIPY Dyes. *Angew. Chem. Int. Ed.*, **2026**, e3726851. [Link]

M.A. Filatov,* T. Mikulchyk, M. Hodée, M. Dvoracek, V.N.K. Mamillapalli, A. Sheehan, C. Newman, S.M. Borisov, D. Escudero, I. Naydenova, Enhancement of Intersystem Crossing in Asymmetrically Substituted BODIPY Photosensitizers. *J. Mater. Chem. C*, **2025**, *13*, 6993-7003. *Highlighted on the front cover.* [Link]

M. Dvoracek, C. Newman, M. Drobizhev, B. Twamley, M.O. Senge, S.A. Vinogradov, M.A. Filatov*, Synthesis and Optical Properties of Unsymmetric Aromatically - Extended BODIPY. *J. Org. Chem.*, **2025**, *90*, 12984–12997. *Highlighted on the front cover.* [Link]

A. Sheehan, I.A. Okkelman, G. Groslambert, C. Bucher, R.I. Dmitriev, M.A. Filatov*, Optoelectronic Properties and Fluorescence Lifetime Imaging Application of Donor-Acceptor Dyads Derived From 2,6-DicarboxyBODIPY. *Chem. Eur. J.*, **2025**, *31*, e202404188. *Highlighted on the front cover.* [Link]

T. Mikulchyk, S. Karuthedath, C. De Castro, A.A. Buglak, A. Sheehan, A. Wieder, F. Laquai, I. Naydenova, M.A. Filatov*, Charge Transfer Mediated Triplet Excited State Formation in Donor-Acceptor-Donor BODIPY: Application for Recording of Holographic Structures in Photopolymerizable Glass, *J. Mater. Chem. C*, **2022**, *10*, 11588-11597. *Highlighted on the back cover.* [Link]

M.A. Filatov*, Heavy-atom-free BODIPY Photosensitizers with Intersystem Crossing Mediated by Intramolecular Photoinduced Electron Transfer. *Org. Biomol. Chem.*, **2020**, *18*, 10-27. *Highly cited paper (>300 citations).* [Link]

M.A. Filatov*, S. Karuthedath, P.M. Polestshuk, H.Savoie, K.J. Flanagan, C. Sy, E. Sitte, M. Telitchko, F. Laquai, R.W. Boyle, M.O. Senge, Generation of Triplet Excited States via Photoinduced Electron Transfer in *meso*-anthra-BODIPY: Fluorogenic Response toward Singlet Oxygen in Solution and *in Vitro*. *J. Am. Chem. Soc.*, **2017**, *139*, 6282−6285. *Highly cited paper (>300 citations).* [Link]

M.A. Filatov, A. Y. Lebedev, S.N. Mukhin, S. A. Vinogradov and A. V. Cheprakov, π-Extended Dipyrrins Capable of Highly Fluorogenic Complexation with Metal Ions. *J. Am. Chem. Soc.*, **2010**, *132*, 9552-9554.[Link]